

Md Monjurul Islam Rifat

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PROFILE SUMMARY

Ph.D. researcher at the University of Florida specializing in the evolution of mutation rates and spectra across *Caenorhabditis* nematodes. Research integrates Mutation Accumulation (MA) experiments with population-genomic analyses to connect experimentally inferred mutational mechanisms to natural patterns of genomic variation. Brings extensive expertise in whole-genome sequencing (WGS), long-read (PacBio HiFi) bioinformatics, large-scale variant calling, and Bayesian statistical modeling of evolutionary processes. Dissertation work spans both short-read and long-read sequencing platforms, with reproducible HPC pipeline development on UF's HiPerGator cluster.

EDUCATION

Ph.D. in Zoology, University of Florida,	2026
<i>Dissertation: " Evolution of the Mutational Process Across Experimental and Natural Populations in <i>Caenorhabditis</i> Nematodes"</i>	
Advisor: Dr. Charles F. Baer	
M.S. in Mathematics, University of Louisiana at Lafayette	2019
B.S. in Statistics, Biostatistics & Informatics, University of Dhaka	2015

RESEARCH INTERESTS

Mutation rate and spectrum evolution; Comparative population genomics; Recombination and genetic variation; Long-read sequencing bioinformatics; Bayesian statistical modeling of evolutionary processes; Natural selection and drift in shaping genome-wide mutational patterns

REPRESENTATIVE PUBLICATIONS

Peer-Reviewed Articles

1. **Saber, S. *, Rifat, M.M.I. ***, Rahimi, F., et al. (2026). Evolution of the rate, spectrum, and fitness effects of mutation under minimal selection in *Caenorhabditis elegans*. *Genetics*, 233(1); iyag085. (***Co-first author**)
2. Saber, S., Johnson, L.M., Rifat, M.M.I., et al. (2025). Cumulative effects of mutation and selection on susceptibility to bacterial pathogens in *Caenorhabditis elegans*. *Evolution*, 79(10), 2047–2056.
3. Kim, S., & Rifat, M.M.I. (2021). Diagnostic analysis of a circular–circular regression model using asymmetric and/or bimodal circular errors. *Communications in Statistics: Theory and Methods*, 50(12), 2848–2858.

Manuscripts in Preparation

4. Rifat, M.M.I., et al. (in preparation). Evolution of the mutational process in *Caenorhabditis*: new results from *C. briggsae*.
5. Rifat, M.M.I., et al. (in preparation). Variation in the nucleotide sequence spectra in the genus *Caenorhabditis*: Implications for the evolution of the mutational process.

CONFERENCE PRESENTATIONS

Oral Presentations

- Rifat, M.M.I.** et al. Evolution of the Mutational Process in *Caenorhabditis*: New Results from *C. briggsae*. Annual Evolution Meeting 2026, Cleveland, OH. (2026)
- Rifat, M.M.I.** et al. New Insights on the Evolution on the of the Mutation Rate and Spectrum in Nematodes using *Caenorhabditis briggsae* Mutation Accumulation Lines. 10th Annual Florida Worm Meeting, Florida Institute of Technology, Melbourne, FL. (2026)
- Rifat, M.M.I.** et al. Excursions along the Drift Barrier: Evolution of the Rate and Spectrum of Spontaneous Mutations in *C. elegans* under Minimal Selection. Southeastern Population Ecology & Evolutionary Genetics Meeting, Great Smoky Mountain Institute, Tremont, TN. (2025)
- Rifat, M.M.I.** et al. Evolution of the Rate and Spectrum of Spontaneous Mutations in *C. elegans* under Relaxed Selection. Annual Evolution Meeting 2025, Athens, GA. (2025)
- Rifat, M.M.I.** et al. Evolution of the *C. elegans* Mutational Process under Relaxed Selection. 9th Annual Florida Worm Meeting, Florida Institute of Technology, Melbourne, FL. (2025)

Poster Presentations

- Rifat, M.M.I.** et al. Variation in the Nucleotide Sequence Spectra in the Genus *Caenorhabditis*: Implications for Evolution. University of Florida, Gainesville, FL. (2024)
- Rifat, M.M.I.** et al. An Empirical Investigation on the Evolution of the *C. elegans* Mutation Rate and Spectrum in Real Time. Clemson University, Clemson, SC. (2024)

Departmental & Institutional Presentations

- Rifat, M.M.I.** Evolution of the Mutation Rate and Spectra in *Caenorhabditis elegans*. 3-Minute Thesis (3MT) Competition, University of Florida, Gainesville, FL. (2025)
- Rifat, M.M.I.** et al. Excursions along the Drift Barrier: Evolution of the Rate and Spectrum of Spontaneous Mutations in *C. elegans* under Minimal Selection. Worm Club, University of Florida, Gainesville, FL. (2025)

FELLOWSHIPS & AWARDS

Grinter Fellowship (\$6,000 total; \$2,000/year), University of Florida 2020–2023

RESEARCH SKILLS

Genomic & Bioinformatic Analysis

- Whole-genome sequencing (WGS): genome assembly, annotation, large-scale variant calling, and downstream analyses
- Long-read sequencing (PacBio HiFi): end-to-end pipeline development for long-read MA experiments
- Variant calling tools: GATK (HaplotypeCaller, GenomicsDBImport), DeepVariant, GLnexus, bcftools, samtools, BEDTools, mosdepth, SnpEff
- Quality control: false positive/negative filtering, contamination analysis, mapping error detection
- HPC pipeline development and SLURM job scheduling on UF HiPerGator cluster
- Analysis of SNVs, indels, MNMs, and tandem repeats from WGS data

Statistical Modeling & Inference

- Bayesian MCMC and likelihood-based inference for evolutionary and population-genetic models
- Non-negative Matrix Factorization (NMF) for mutational signature decomposition
- Compositional data analysis of nucleotide substitution spectra
- Hierarchical mixed-effects models for mutation rate and spectrum estimation

- Bootstrap, permutation, and simulation-based model validation

Programming & Computational Tools

- R (primary): tidyverse, vcfR, ggplot2, lme4, rstan; Python: biopython, pandas, numpy; Bash/Shell scripting
- Reproducible pipeline development for high-dimensional genomic data analysis

Model Organisms & Datasets

- *Caenorhabditis elegans*: MA lines from N2 (Bristol), CB4856 (Hawaiian), and PB306
- *Caenorhabditis briggsae*: MA lines from HK104 (Japan) and PB800 (Ohio); short-read and long-read (HiFi) datasets
- *Caenorhabditis brenneri*: MA lines from CFB2252 strain (work in progress)
- Wild isolates of *C. elegans*, *C. briggsae*, and *C. tropicalis* from the CaENDR database

UNDERGRADUATE MENTORSHIP

Newhall, Yell — University of Florida

2024–2025

Supervised undergraduate research on genome assembly, variant calling, and mutation identification using whole-genome sequencing data and bioinformatics pipelines.

TEACHING EXPERIENCE

Teaching Assistant, University of Florida 2020–Present

BSC2011L: Integrated Principles of Biology Lab 2 — Summers 2023, 2025 and 2026

PCB3063: Genetics — Fall 2023, 2024, 2025, Spring 2023, 2024, 2025 and 2026

BSC2005L: Lab in Biological Sciences — Summer 2024

BSC2005: Biology for non-Majors — Fall 2021, 2022, Summer 2022

BSC2010L: Integrated Principles of Biology Lab 1 — Fall 2020, Spring 2021, 2022, Summer 2021

Teaching Assistant, University of Louisiana at Lafayette 2016–2017

Instructor, University of Louisiana at Lafayette 2017–2019

Stat 214: Elementary Statistics — 2018–2019

Math 103–104: College Algebra — 2017

PROFESSIONAL MEMBERSHIPS

Society for the Study of Evolution (SSE)

American Society of Naturalists (ASN)

SERVICE & LEADERSHIP

Pre-Grad Poster Award Judge: 10th Annual Worm Meeting, 2026

Faculty Recruitment Committee: Marine Biodiversity Search 2023–2024

BGSA (Biology Graduate Student Association) Graduate Representative, UF: 2023–2024

BGSA Climate Committee Member: 2022–2023

BGSA Committee: Public Relations & Fundraising Officer: 2021–2022

BSA (Bangladesh Student Association), ULL, Lafayette: Treasurer: 2017–18

LANGUAGE PROFICIENCY

Bengali (Native) • English (Fluent)

REFERENCES

Dr. Charles F. Baer — Professor, Department of Biology, University of Florida (Dissertation Advisor)
cbaer@ufl.edu

Dr. Juannan Zhou — Assistant Professor, Department of Biology, University of Florida (Dissertation Committee & Collaborator)
juannanzhou@ufl.edu

Dr. Moein Rajaei — Postdoctoral Associate, Yale University (Collaborator)
moein.rajaei@yale.edu